

CS & IT ENGINEERING



Computer Network - 1

Transport Layer

Lecture No. – 04

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Recap of Previous Lecture



Topic

Connection-Oriented Demultiplexing

Topic

TCP Header



Topics to be Covered



Topic

TCP Connection Establishment

Topic

TCP Data Transfer





Topic : TCP Operation



Three phases of TCP operation :

1. Connection establishment

→ 3-way handshake process that establishes a connection

2. Data transfer

3. Connection termination

→ 4-way handshake process that closes the connection



Topic : TCP Connection Establishment



- Connection establishment between TCP client and TCP server
- 3-way handshake process
- Always TCP client initiate the connection request to TCP server

Client_ISN = X

TCP Client

TCP Server

Server_ISN = Y



CLOSE ←

Connection Req.

Seq_no = X , ACK_no = G
SYN_flag = 1, ACK_flag = 0
No Payload

[SYN_SENT]

[LISTEN]

Connection Accept
Seq_no = Y , ACK_no = $(X+1)$
SYN_flag = 1, ACK_flag = 1
No payload

[SYN_RECV]

FINAL-ACK
Seq_no = $(X+1)$, ACK_no = $(Y+1)$
SYN_flag = 0, ACK_flag = 1
(with or without payload)

[ESTABLISHED]

[ESTABLISHED]



Topic : TCP Connection Establishment



→ Initial Sequence Number : if SYN flag is on [both TCP client and server randomly chooses their initial sequence number, to prevent from some kind of attacks] [ISN]

→ SYN segment do not carry user data [no any payload]

→ SYN segment consume one sequence number ✓

Client_ISN = X

TCP Client

TCP Server

Server_ISN = Y



Seq_no = X
SYN_flag = 1

Seq_no = Y , ACK_no = $(X+1)$
SYN_flag = 1, ACK_flag = 1

FINAL-ACK
Seq_no = $(X+1)$, ACK_no = $(Y+1)$
ACK_flag = 1, (No Payload)

Seq_no = $(X+1)$, ACK_no = $(Y+1)$
ACK_flag = 1, Data in Payload

ESTABLISHED

ESTABLISHED

Client_ISN = X

TCP Client

TCP Server

Server_ISN = Y



Seq_no = X
SYN_flag = 1

Seq_no = Y , ACK_no = $(X+1)$
SYN_flag = 1, ACK_flag = 1

FINA-ACK
Seq_no = $(X+1)$, ACK_no = $(Y+1)$
ACK_flag = 1, (Data in Payload)

Ex:-
HTTP Req.

ESTB

ESTB

[MSQ]

IIT-R

[GATE-2025][1 Mark]



#Q. Consider the 3-way handshaking protocol for TCP connection establishment. Let the three packets exchanged during the connection establishment be denoted as P1, P2, and P3, in order. Which of the following option(s) is/are TRUE with respect to TCP header flags that are set in the packets?

☒ A

P1: SYN = 1

☒ B

P2: SYN = 1, ACK = 1

☐ C

P2: SYN = 0, ACK = 1

☐ D

P3: SYN = 1, ACK = 1

$P_1 = \text{SYN}$

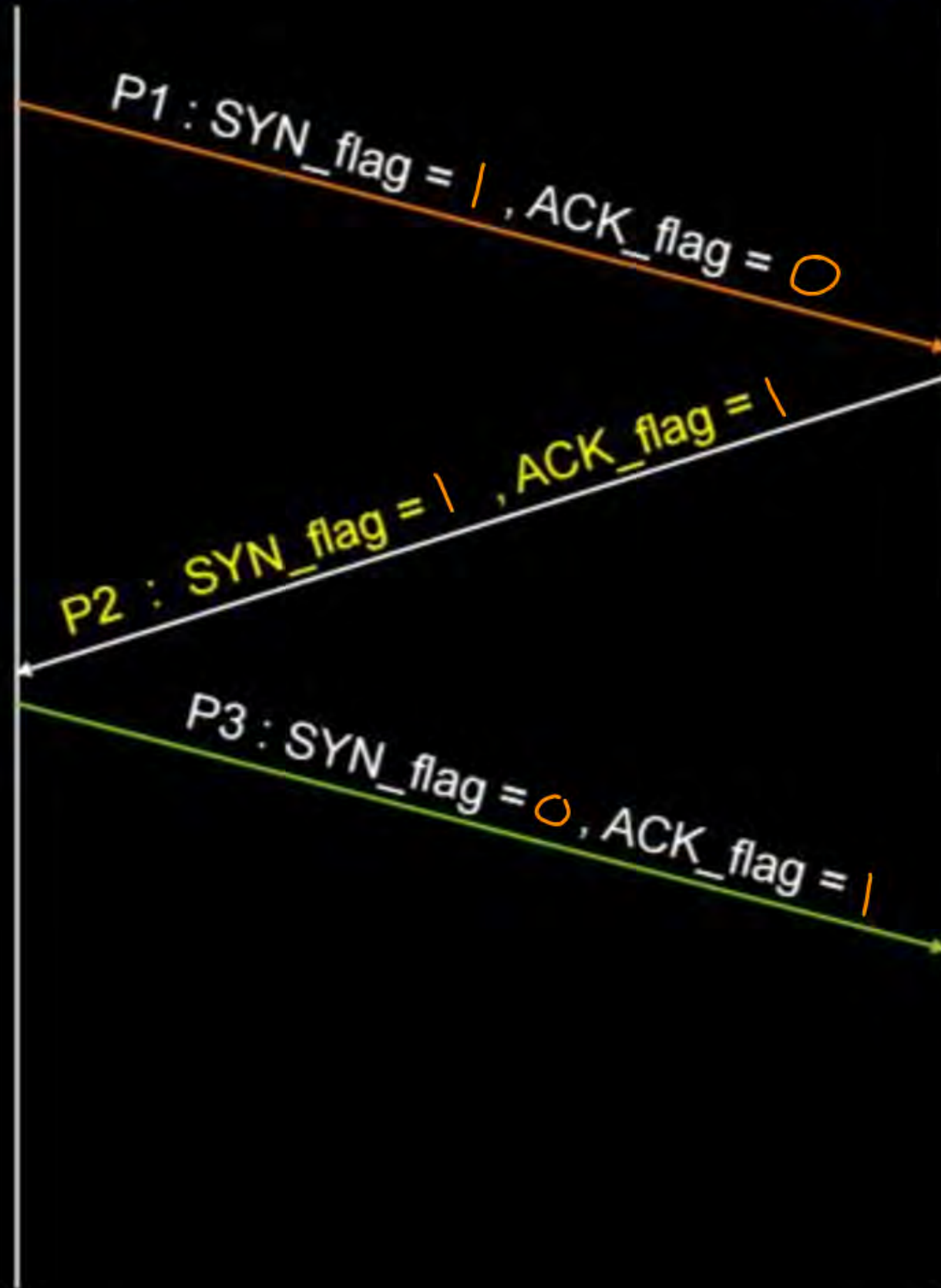
$P_2 = \text{SYN} + \text{ACK}$

$P_3 = \text{ACK}$

Ans: A & B

TCP Client

TCP Server



[MSQ]

115C

[GATE-2024][1 Mark]



#Q. TCP client P successfully establishes a connection to TCP server Q. Let N_P denote the sequence number in the SYN sent from P to Q. Let N_Q denote the acknowledgement number in the SYN ACK from Q to P. Which of the following statements is/are CORRECT?

- ↓
- ☒ A The sequence number N_P is always 0 for a new connection F
 - ☒ B The sequence number N_P is chosen randomly by P T
 - ☒ C The acknowledgement number N_Q is equal to N_P F
 - ☒ D The acknowledgement number N_Q is equal to $N_P + 1$ T

$$N_Q = (N_P + 1)$$

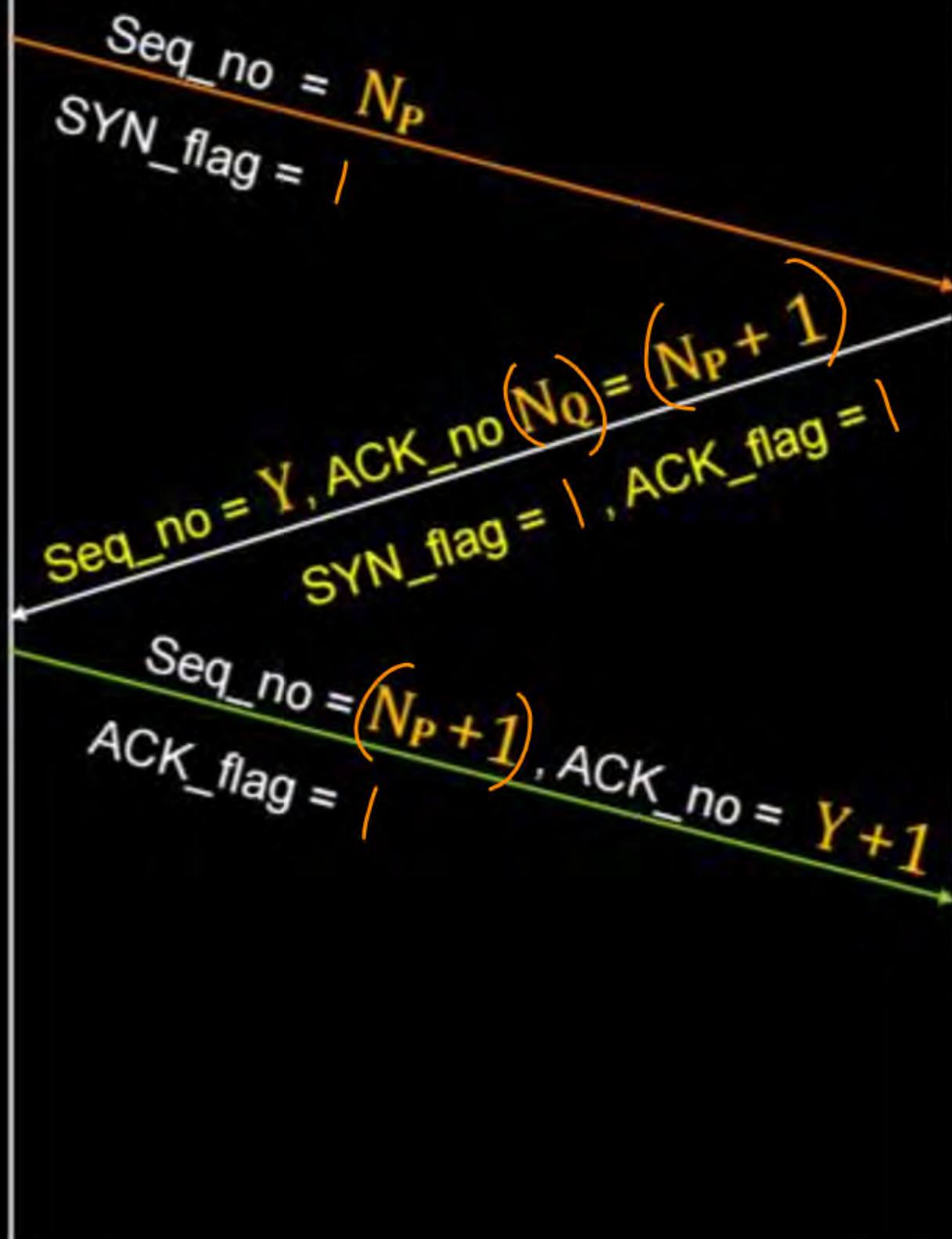
Ans: B & D

Client_ISN = N_P

TCP Client

TCP Server

Server_ISN = Y

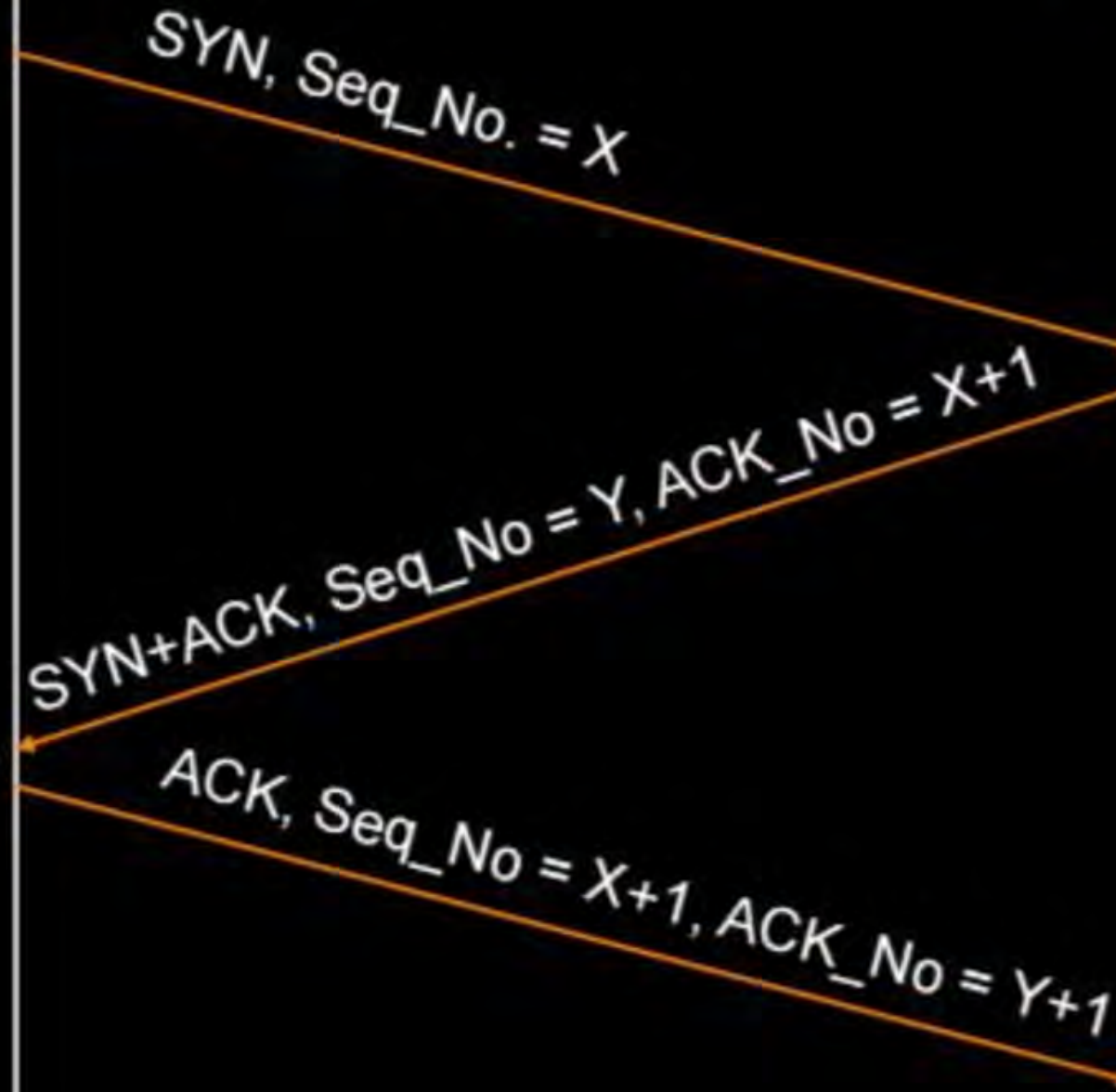


Client_ISN = X

TCP Client

TCP Server

Server_ISN = Y



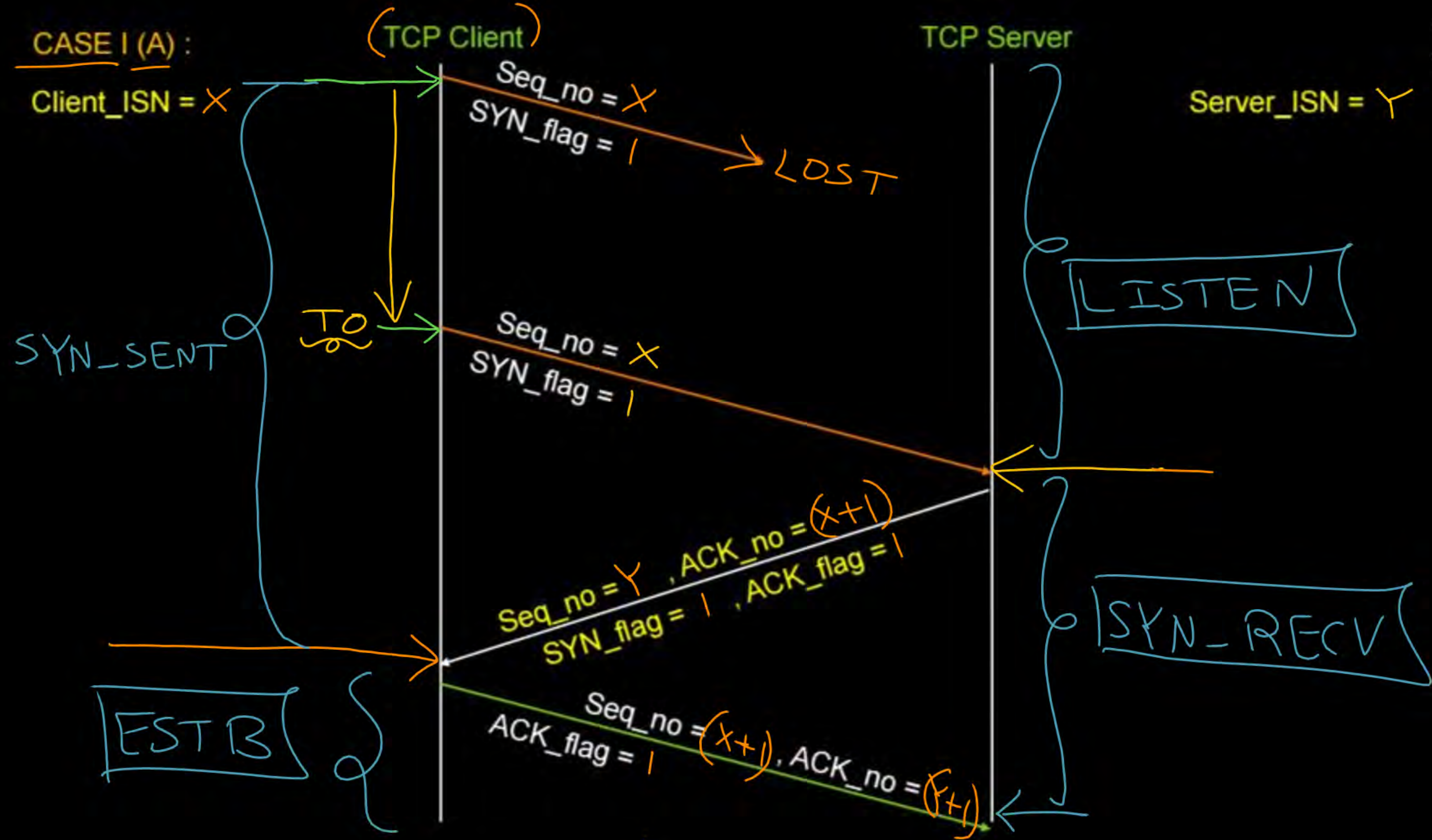
CASE I (A) :

Client_ISN = X

(TCP Client)

TCP Server

Server_ISN = Y



CASE I (B) :

Client_ISN = X

Z

TCP Client

TCP Server

Server_ISN = Y

Seq_no = X
SYN_flag = 1

LOST

T_0

Seq_no = Z
SYN_flag = 1

Seq_no = Y , ACK_no = $(Z+1)$
SYN_flag = 1, ACK_flag = 1

Seq_no = $(Z+1)$, ACK_no = $(Y+1)$
ACK_flag = 1

CASE II (A):

Client_ISN = X

TCP Client

TCP Server

Server_ISN = Y

Seq_no = X
SYN_flag = 1

LISTEN

Seq_no = Y, ACK_no = X+1
SYN_flag = 1, ACK_flag = 1

LOST

To

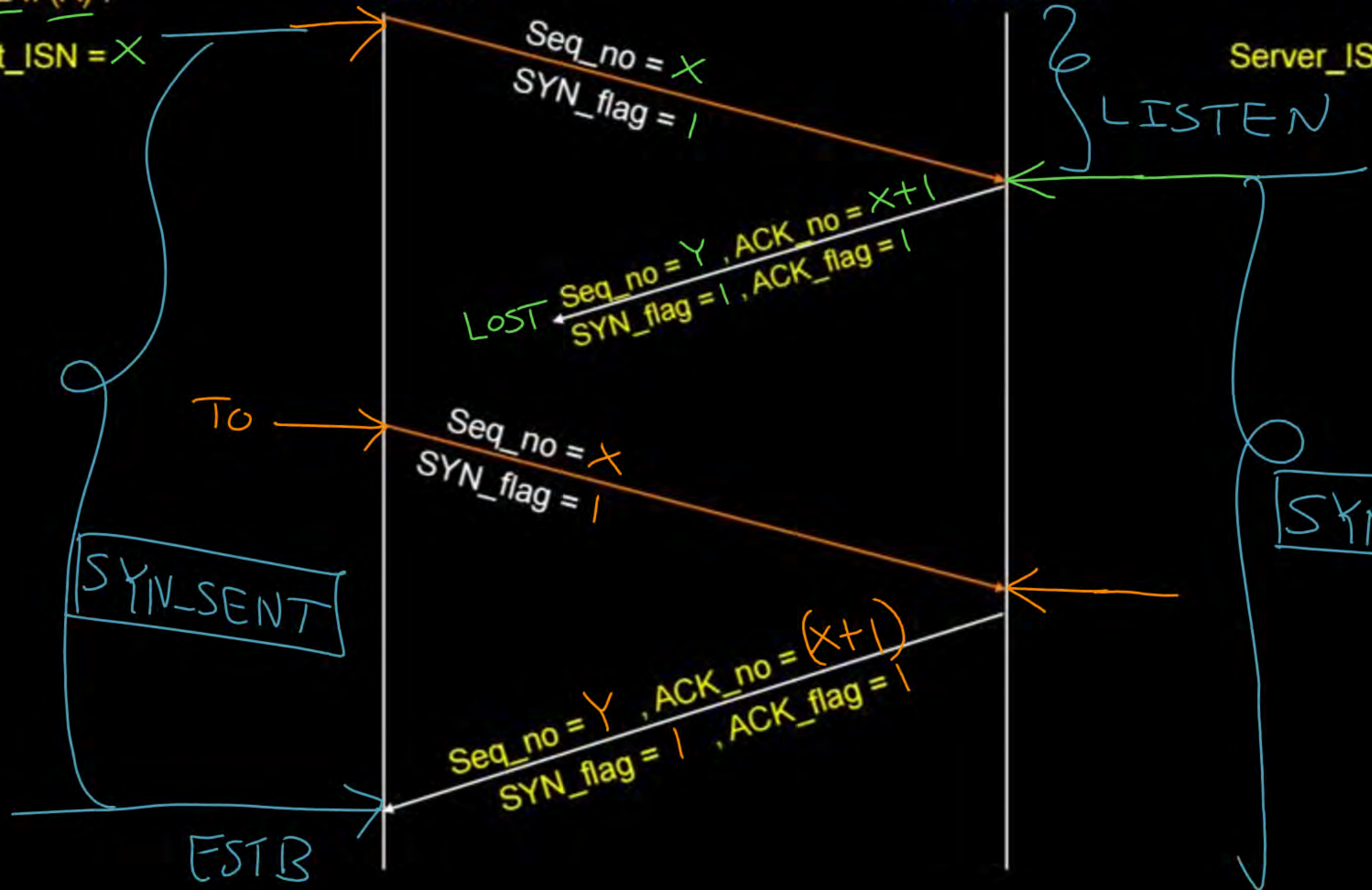
Seq_no = X
SYN_flag = 1

SYN_SENT

SYN_RECV

Seq_no = Y, ACK_no = (X+1)
SYN_flag = 1, ACK_flag = 1

ESTB



CASE II (B) :

Client_ISN = ~~X~~

TCP Client

TCP Server

Server_ISN = Y

LISTEN

Seq_no = X
SYN_flag = 1

Seq_no = Y, ACK_no = X+1
SYN_flag = 1, ACK_flag = 1

Seq_no = Z
SYN_flag = 1

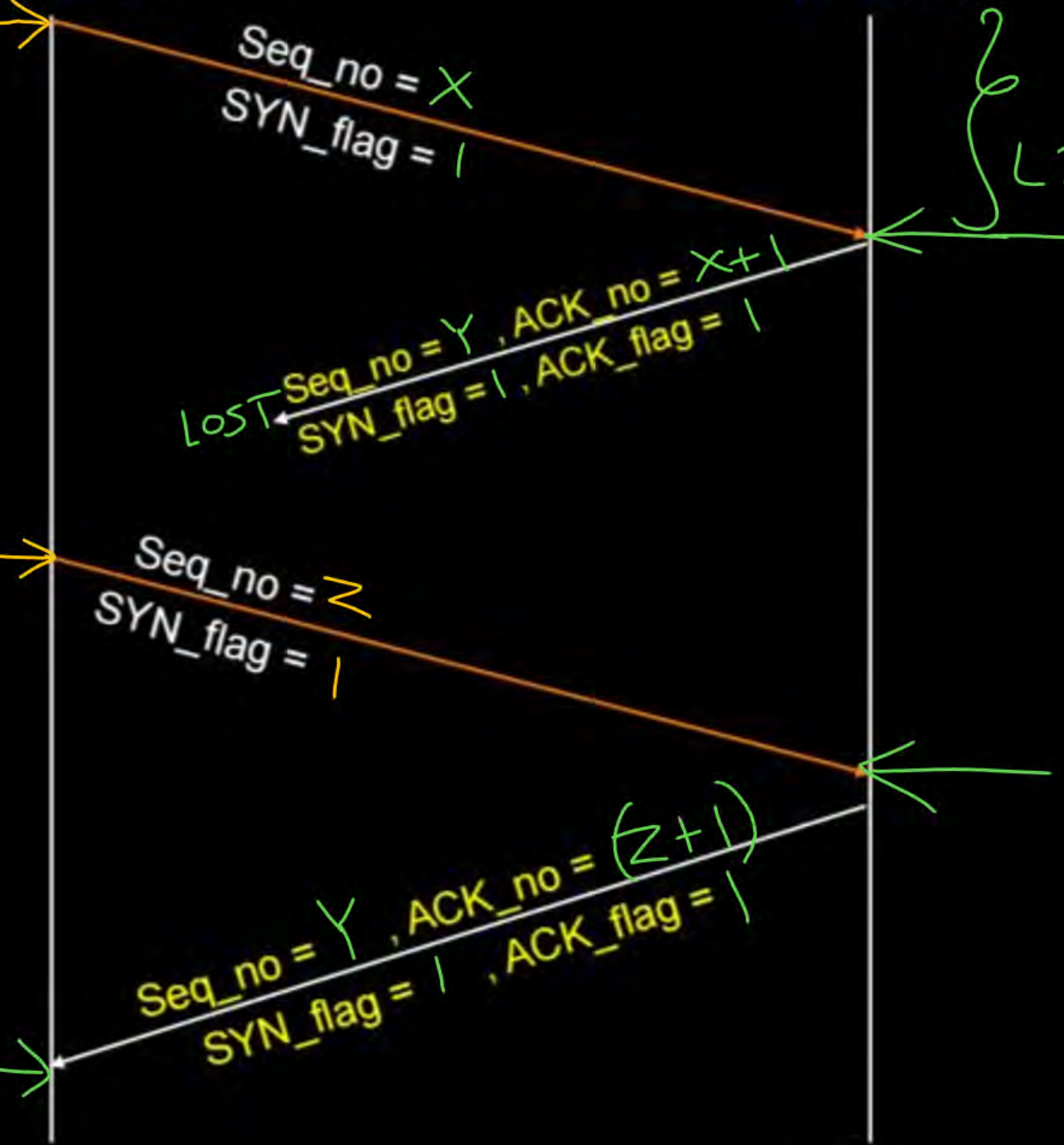
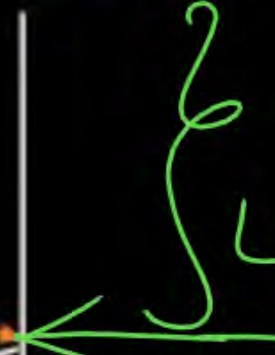
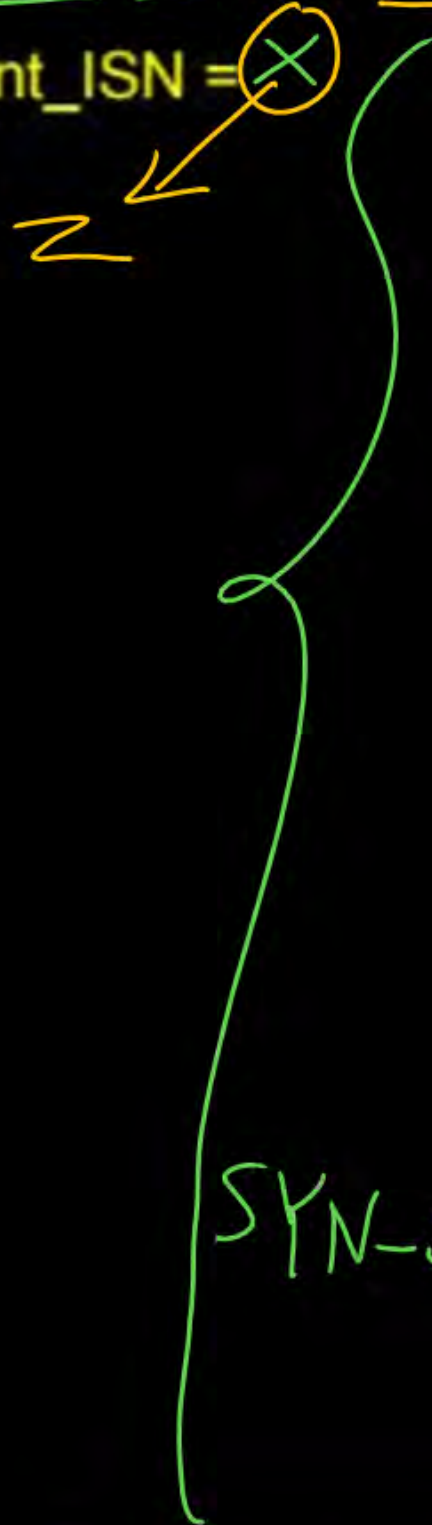
Seq_no = Y, ACK_no = (Z+1)
SYN_flag = 1, ACK_flag = 1

SYN_RECV

SYN-SENT

To

lost



CASE III (A):

Client_ISN = X

TCP Client

TCP Server

Server_ISN = Y

SYN_SENT

TO

ESTB

No data sent

SYN_RECV

ESTB

Seq_no = X
SYN_flag = 1

LISTEN

Seq_no = Y , ACK_no = $(X+1)$
SYN_flag = 1, ACK_flag = 1

Seq_no = X
SYN_flag = 1
Seq_no = $X+1$, ACK_no = $Y+1$
ACK_flag = 1

Seq_no = Y , ACK_no = $X+1$
SYN_flag = 1, ACK_flag = 1

Seq_no = $X+1$, ACK_no = $Y+1$
ACK_flag = 1

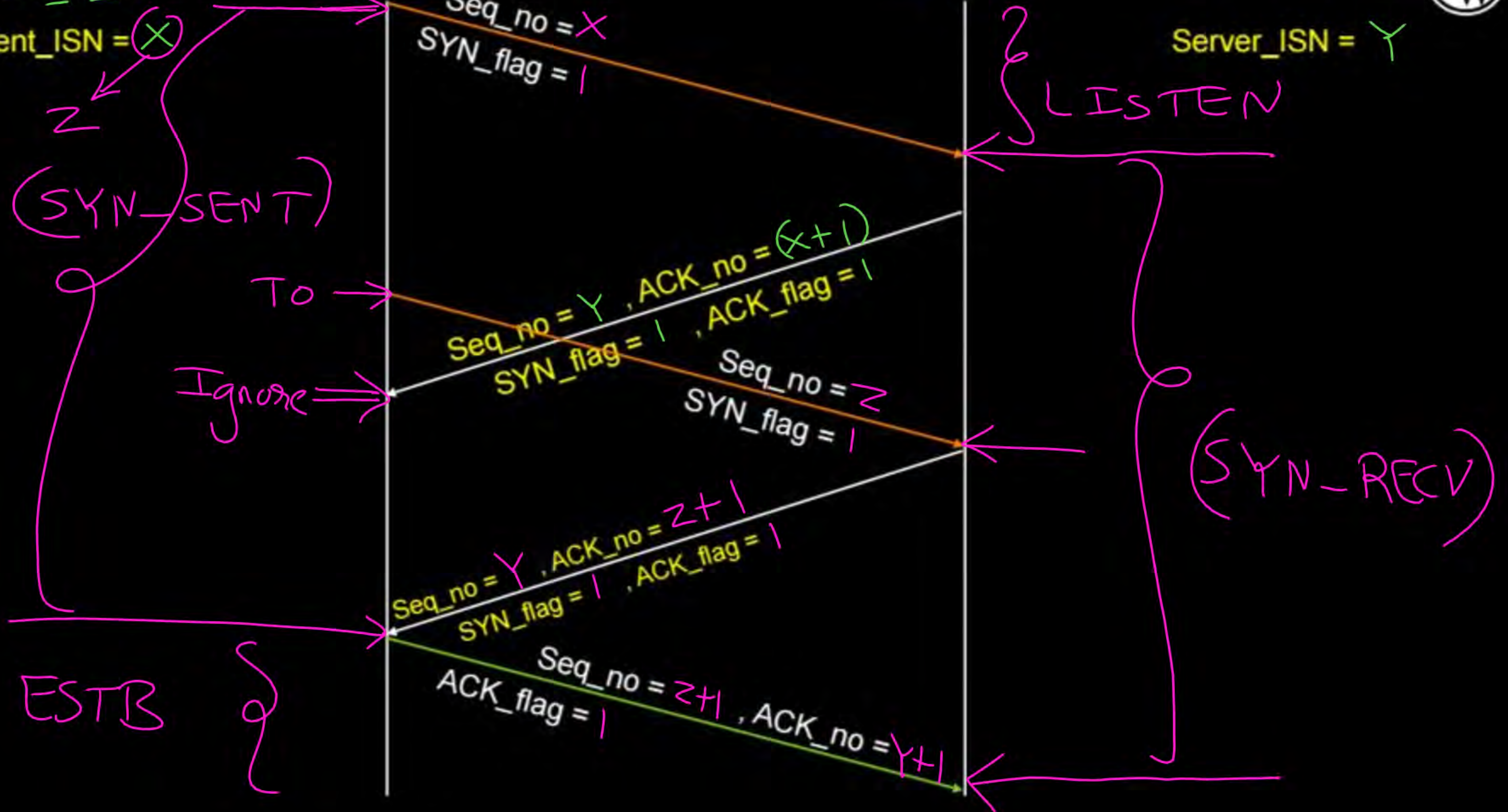
CASE III (B) :

Client_ISN = $\odot$

TCP Client

TCP Server

Server_ISN = γ



CASE IV:

Client_ISN = X

TCP Client

TCP Server

Server_ISN = Y

SYN_SENT

ESTB

(SYN_RECV)

ESTB

Seq_no = X
SYN_flag = 1

Seq_no = Y, ACK_no = X+1
SYN_flag = 1, ACK_flag = 1

FINAL-ACK
Seq_no = X+1, ACK_no = Y+1
ACK_flag = 1, (No Payload) LOST

Seq_no = (X+1), ACK_no = Y+1
ACK_flag = 1, Data in Payload

CASE V:

Client_ISN = X

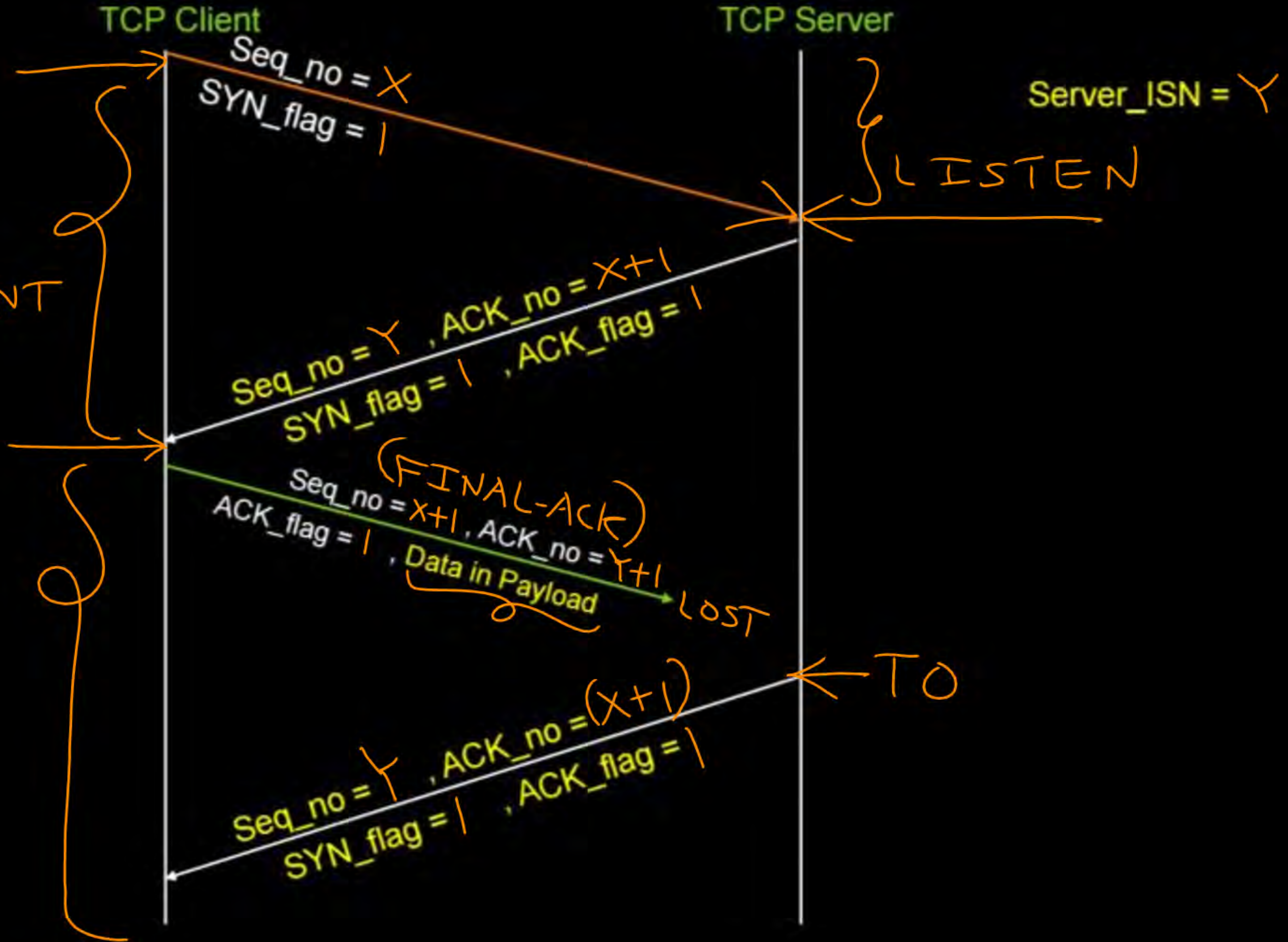
TCP Client

TCP Server

Server_ISN = Y

SYN-SENT

ESTB



[MCQ]

11SC, H.W

[GATE-IT-2008] [1 Mark]



#Q. Which of the following statements are TRUE for 'three way handshake' for TCP connection establishment ?

(S1) Loss of SYN + ACK from the server will not establish a connection

(S2) Loss of ACK from the client cannot establish the connection

(S3) The server moves LISTEN → SYN_RCVD → SYN_SENT → ESTABLISHED in the state machine on no packet loss

(S4) The server moves LISTEN → SYN_RCVD → ESTABLISHED in the state machine on no packet loss

A S2 and S3 only

B S1 and S4 only

C S1 and S3 only

D S2 and S4 only



2 mins Summary



Topic

TCP Connection Establishment ✓

Topic

~~TCP Data Transfer~~



THANK - YOU

